	CUI CAPS – Demographics – Clinical Safety Case and Closure Report			
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DEPARTMENT OF HEALTH INFORMATICS DIRECTORATE

Common User Interface (CUI) Programme

Clinical Applications and Patient Safety (CAPS) Project:

Demographics / Information Input and Display:

Patient Banner (Patient Identification)

Address Input and Display

NHS Number Input and Display

Patient Name Input and Display

Sex and Current Gender Input and Display

Telephone Number Input and Display

Date Display

Time Display

Date and Time Input

Clinical Safety Case and Closure Report

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Glossary of Terms

Acronym / Term	Definition
Accredited Clinician	Those clinicians that have attained accreditation status under the DHID / NPSA Accreditation Training Scheme for clinical risk management.
ALARP	As low as reasonably practicable.
CAPS	Clinical Applications and Patient Safety Project.
CATR	Clinical Authority to Release means a certificate issued by the DHID Clinical Safety Group (CSG) to indicate that it has completed the process of reviewing the submitter's evidence intended to substantiate the assertion that a system is clinically safe and thereby fit for purpose.
Clinical Safety	Clinical Safety means the process of reviewing and dealing appropriately with Clinical Hazards and Clinical Risks in order to ensure patient safety, including anything which has the potential to cause harm to a patient, but specifically excluding health and safety considerations in terms of operating the Services or systems.
CSA	The CSA (Clinical Safety Advisor) is the nominated DHID individual, who is also an Accredited Clinician, responsible for the management of clinical safety in the CUI Programme.
CSC	Clinical Safety Case.
CSG	Clinical Safety Group (DHID) – responsible for awarding CATR.
CSMS	Clinical Safety Management System (DHID) – the DHID prescribed process for management of clinical safety.
CUI	Common User Interface Programme.
ISB	Information Standards Board for Health and Social Care.
DHID	Department of Health Informatics Directorate.
NPSA	National Patient Safety Agency.
PSA	Patient Safety Assessment.
SCR	Safety Closure Report.

1. Introduction

This document is the combined Clinical Safety Case (CSC) and Safety Closure Report (SCR) for the following related set of Common User Interface (CUI) design guides:

- Patient Banner
- Address Input and Display
- NHS Number Input and Display
- Patient Name Input and Display
- Sex and Current Gender Input and Display
- Telephone Number Input and Display
- Date Display
- Time Display
- Date and Time Input.

This report is part of a set of clinical safety documentation, which has been produced in order to meet requirements of the Department of Health Informatics Directorate (DHID) Clinical Safety Management System (CSMS). It contains the Hazard Logs and supporting development information for above 9 design guides. It is being submitted to the DHID Clinical Safety Group (CSG) for review and sign off in order to achieve Clinical Authority to Release (CATR).

The audience for this document is the DHID CSG, DHID National Integration Centre (NIC), Information Standards Board for Health and Social Care (ISB), the CUI Clinical Applications and Patient Safety (CAPS) team, clinical / healthcare system suppliers and health organisations.

2. Background

Established in 2005, the Common User Interface (CUI) programme within the (recently re-organised) Department of Health Informatics Directorate Technology Office is focused on supporting the NHS, via delivery of a series of projects, to:

- Increase patient safety;
- Increase clinician effectiveness;
- Make IT systems easier to use;
- Maintain a well managed and more secure desktop and server infrastructure; and
- Enable improved personal and team productivity.

The Clinical Applications and Patient Safety (CAPS) project, part of the CUI programme, is responsible for the development of platform agnostic, text based user interface design guidelines for healthcare IT suppliers to the NHS. The scope of the CAPS project is focused on those patient safety critical aspects of clinical systems (such as medications management and patient identification) in order to increase consistency and thereby improve the safety integrity of the many different clinical systems which will be used throughout the NHS.

Since December 2005, the CAPS project has and continues to:

- Deliver a growing portfolio of NHS driven clinical applications design guidance;
- Engage, involve and improve awareness in the NHS clinical community of their ability to drive change in clinical supplier UI;

Support other programmes of activity within DHID in their UI needs, where applicable to the CAPS roadmap;

- Develop and maintain good working relationships with and encourage adoption by NHS clinical application suppliers;
- Develop and implement the CUI CAPS Clinical Safety Management Plan – in accordance with the DHID Clinical Safety Management System.
- Progress published guidance as potential information standards for the NHS in England through the Information Standards Board for Health and Social Care (ISB);
- Engage with the DHID National Integration Centre (NIC) technical assurance process;
- Foster and maintain an awareness of the project's objectives with both international clinical and supplier communities.

3. CUI CAPS Clinical Safety Management System

The following section outlines the *high level* CUI CAPS Clinical Safety Management System – which is predominantly based on the DHID Clinical Safety Management System and the emerging information standard, "Application of Clinical Risk Management to the Manufacture of Health Software." The individual, tailored clinical safety management plan for each deliverable, ie. design guide, covered by this report can be found at Section 6 – Patient Safety Assessment.

3.1. Structure of the CUI CAPS Clinical Safety Team

The list below identifies the key people responsible for clinical safety within the CUI CAPS project. Although, all members of the CUI CAPS team are expected to integrate clinical safety into every aspect of development work. Appropriate awareness raising and education has been provided 'on site' by the current Clinical Safety Advisor. Relevant clinical team members of the CUI CAPS team (known as the Specific Audience) have attended the DHID training in human factors and risk assessment, in order to become "Accredited Clinicians" in Patient Safety Assessment (PSA) workshops and understand the required safety processes.

- Beverley Scott, Clinical Safety Advisor and Registered Mental Health Nurse (RMN) (Accredited Clinician) – February 2008 – December 2009.
- Lindsey Butler, Clinical Safety Advisor and Registered General Nurse (RGN) (Accredited Clinician) – from August 2009 onwards.
- Stephen Corbett, Head of User Interface Design, Technology Office.

- Tim Chearman, CUI CAPS Project Lead and User Experience Designer.
- Frank Cross, Consultant Surgeon and Senior Clinical Advisor (Accredited Clinician).
- Pete Johnson, Clinical Architect and General Practitioner (Accredited Clinician).

Due to the size of the CUI CAPS team, a internal clinical safety *committee* is not required. Instead the CAPS safety team as defined above meets on a regular basis with the design team to review the CUI CAPS clinical safety approach, individual Hazard Logs and clinical safety products in production and planned for future releases.

The CUI CAPS team utilise the safety assurance and sign off processes of the DHID Clinical Safety Group (CSG) and the Information Standards Board for Health and Social Care (ISB) for CUI design guidance.

3.2. Clinical Safety Products and Sign Off

The following clinical safety specific products will be produced for each CUI CAPS design guide:

- Hazard Log.
- Clinical Safety Case (including the Patient Safety Assessment).
- Safety Closure Report.

The DHID CSMS supports staged submission of the above documents to the Clinical Safety Group as a usual route to achieving Clinical Authority to Release (CATR). However, due to tight CUI CAPS project timescales, ie. typically 3 months per design guide, and limited Clinical Safety Advisor resource, the CUI CAPS Clinical Safety Cases and Safety Closure Reports will normally be compiled and submitted post design guide development.

Depending on the size and complexity of the design guide under consideration, project timescales and agreement from the CSG, the Clinical Safety Case and Safety Closure Report may be combined into one document and one submission to the CSG (as in this instance).

Whilst design guides may be published on the CUI website, www.cui.nhs.uk (registration required) before receiving CATR, both the safety and information standards status of each design guide is clearly identified on the website, and a caveat included in the document itself with respect to the patient safety process.

The latest safety information can be requested by suppliers by emailing the mailbox cuistakeholder.mailbox@nhs.net

3.3. Clinical Safety Risk Management File

All documents created as part of the CUI CAPS Clinical Safety Management System are maintained, version controlled and managed by the CUI CAPS Clinical Safety Advisor (CSA).

Documents are stored on the CUI CAPS Clinical Safety SharePoint site, which has access control measures in place. Members of the CSG or ISB can request access by contacting the CUI CAPS CSA.

3.4. CUI CAPS Clinical Safety Approach

The CUI CAPS project has developed their clinical safety management approach based on the Clinical Safety Management System (CSMS) prescribed to system suppliers by the DHID and the emerging information standard, "Application of Clinical Risk Management to the Manufacture of Health Software."

Although outputs from the CUI CAPS project, namely text based user interface design guidance, are not systems per se, they will inform the user interface work carried out by clinical system suppliers which will in turn end up being part of a live deployed system. It is crucial, therefore, that the outputs from CUI CAPS are also clinically safe.

The DHID CSMS outlines four risk areas: End to end clinical process risk; Technical risk; Message risk; and Patient safety risk. As the CAPS project is focused on the user interface layer of an application, the scope of the CAPS safety process is limited to patient safety risks. The other risk types are classed out of CUI scope. However, where related risks are raised, CUI CAPS will record and transfer those risks to the appropriate parties.

As the CUI CAPS Project is responsible for developing user interface design guidelines which will be utilised within clinical applications - not the development of applications themselves – the CUI CAPS Clinical Safety Cases are developed as Pre-Operational. Taking a typical software development lifecycle (see the table below), the scope of CUI design guidance covers the stages of Define Requirements, Design and Develop.

As such, the CUI CAPS Clinical Safety Cases will cover these stages also (although it important to note that the Patient Safety Assessment's themselves are very much focused upon context of use). It is the intention that the CUI CAPS safety documentation will be the starting point for NHS clinical application developers upon which to build their own Clinical Safety Case and, ultimately, pass on to end users within the live estate.

Manufacturers	Define Requirements Design Develop Build Test Release
End Users	Deploy Use Decommission

In order to build the Clinical Safety Case, each CUI CAPS design guide will undergo an end to end Patient Safety Assessment (PSA). It is important to note that a Patient Safety Assessment (PSA) is not a distinct activity, scheduled into the development lifecycle as a milestone. A PSA is the term used to describe the end to end, continuous, iterative assessment to establish the safety integrity of the deliverable. It is integrated into the development lifecycle for each CAPS deliverable right from the outset.

A CUI CAPS PSA consists of two main 'processes': Hazard identification and risk assessment; and Hazard analysis and risk management. Various safety specific activities (eg. risk assessment workshop), methods (eg. Structured What If Technique, Hierarchical Task Analysis) and tools (eg. hazard checklists, cause and effect (Fishbone) diagrams) can be used during a PSA. However, all usual activities undertaken as part of normal research and development processes for CUI CAPS, ie. non safety specific, will pro-actively elicit patient safety issues and hazards on an ongoing basis. These are then fed into the master Hazard Log via the Clinical Safety Advisor for management.

As part of an end to end Patient Safety Assessment (PSA), the following **safety specific** activities should take place for each CUI CAPS deliverable (design guide):

- Each CAPS deliverable should have an assessment during the scoping phase to identify the minimum number and type of *safety specific* activities to take place and when.
- As a minimum, each CAPS deliverable should have:
    - A web based survey, including safety specific questions.
    - One to one interviews with representative end users, including safety specific questions.
    - Patient Safety Risk Assessment (PSA) workshops, eg.

- 1 x Opening Risk Assessment workshop with representative end users (pre design: safety requirements and objectives and preliminary hazard identification and risk assessment).
- 1 x Closing Risk Assessment workshop with representative end users (post design: risk identification, grading and management).
- Interim Risk Assessments can take place as required, such as when a design guide spans multiple releases.

▪ The purpose of an Opening Risk Assessment is to inform the CUI CAPS requirements process (including safety requirements). Very often, the ORA can raise hazards and issues that are either out of scope or not patient safety specific – as it is conducted during the very early, conceptual stages of design.

▪ All hazards and issues, irrespective of where and when they are raised, go into the Hazard Log for central tracking by the CSA.

▪ The purpose of a Closing (or Interim) Risk Assessment is to formally risk assess final (or near final) designs, using a structured approach to identify hazards and grade and manage risks.

▪ Regular meetings take place with the CUI CAPS safety and design teams to review live hazards and manage to an acceptable level.

▪ **The number and type of activities, methods and tools varies across CUI CAPS deliverables (see Section 6 – Patient Safety Assessment for the *actual* activities). <u>Ultimately, the overall aim of the PSA is to build a progressive Clinical Safety Case</u>.**

4. Customer Need and Scope

Currently, clinical systems used within the NHS in England, across all care settings, differ in the way demographic information and dates and times are displayed and input. Inherent within this is the risk that healthcare professionals moving between clinical systems made by different developers can misinterpret information, potentially leading to Patient Safety Incidents (as defined by the National Patient Safety Agency (NPSA)). It is also vital that data is entered into system accurately.

4.1. Patient Banner

The term “patient banner” refers to the area of the user interface that contains header information for a patient record. The patient banner is the area within the user interface that is most often used to match records with patients and contains key information for identifying patients, such as name, address and date of birth. This information may be matched with that provided by the patient (or a third party, such as a parent), usually in their presence or over the telephone. Information in the patient banner may also be matched with that in any of the patient’s associated artefacts, such as samples, letters, specimens, wristbands and x-rays.

The National Patient Safety Agency (NPSA) states that although there are no accurate figures on the frequency or cost of mismatching errors, they form a significant part of the whole range of errors in healthcare. However, there is considerable variation in the layout of a patient banner across clinical applications.

The aims of the Patient Banner described in the CUI design guide are to:

- Ensure patients are correctly identified and matched with their patient record by displaying data items consistently.

- Allow quick access to and display of other summary information, such as contact details and allergies, for a patient.

- Reduce and, where possible, eliminate errors in the matching of patient with their care.

When errors do occur, they are typically one of the three main types:

1. The patient is given the wrong treatment as a result of a failure to identify the patient correctly.

2. The patient is given the wrong treatment as a result of a failure to match the patient correctly with their artefacts (samples, letters, specimens, x-rays, and so on).

3. The patient is given the wrong treatment as a result of a failure in communication between staff, or staff not performing checking procedures correctly.

The design guidance is concerned with each of the errors above, but particularly 1) and 2).

The following items are in scope:

- **Defining the minimum items of information that must be contained in a Patient Banner in order to identify the patient**, such as, but not limited to:
  - Patient name (family name, given name and title)
  - Patient address
  - Date of birth
  - Patient age
  - Date of death (displayed to aid the identification of the deceased patient)
  - Gender
  - NHS Number
- **Grouping and layout of this information**, including:
  - Which data items should be grouped together
  - Which data items should be given the most prominence
  - Whether and where controls (such as expandable / collapsible panels) should be used

- **Labelling of information**, including:
  - Which items of information in the Patient Banner are to be labelled
  - How items of information are to be labelled; this will cover the label text, positioning and any elements of styling required to differentiate labels visually from data
- **Location and shape of the Patient Banner**, in order to achieve:
  - Optimal visibility of the key information needed for patient identification
  - Easy recognition of the Patient Banner in the context of the wider clinical application
- **Size of the Patient Banner**. This must be a balance between the minimum required for suitable layout of the content, and the maximum that can be used at the expense of screen space left for the clinical application to use.

The aspects that are **out of scope** for this standard are:

- **Reduced-size form factors –** This standard does not cover reduced-size form factors, such as personal digital assistants (PDA's) and other small mobile devices.
- **Display of components with separate display standards –** Some individual components in the banner already have separate display standards.
- **Input of information** – The standard only covers the display of the patient banner and its contents. It does not cover the input of any information.
- **Data storage** – The standard relates only to the human interface layer of a software application, and does not prescribe the format for storing patient banner data.
- **Display styles –** The standard does not cover the choice of display features such as font, size, background an foreground text colour.

4.2. Input and Display of Demographic Information

These 5 design guides will specify common user interface aspects of:

- Address Input and Display
- NHS Number Input and Display
- Patient Name Input and Display
- Sex and Current Gender Input and Display
- Telephone Number Input and Display.

The purpose of above design guidance is to make display of patient demographic information consistent across all clinical systems within the NHS in England. The objective is to increase patient safety by maximising clinical utility and minimising reading, input and transcribing errors.

Displaying and entering unambiguous information in a consistent format is a core element in ensuring effective patient care. It is vital that healthcare professionals correctly interpret names, telephone numbers, and other information relating to patient demographics, clinical episodes and planned treatments, among others. It is also vital that data is entered into systems accurately.

Research quantifying the clinical safety of ambiguous or inconsistent display of demographic information is limited, but there is a clear argument that unambiguous and consistent user interfaces would reduce the potential for human error leading to patient safety incidents. This view is supported by the NPSA. For example:

- A misinterpreted address could result in a patient not being informed of an appointment.
- An ambiguous name such as "David Clare", where the order of elements is not obvious, could result in identifying a wrong patient, or the wrong gender of the patient.
- An application that uses Sex and Current Gender labels the opposite way around to another application, which is possible with two such similar fields as these, can result in clinicians that move between the applications making mistakes.

The following items are **in scope**:

- **Address Input and Display:**
- o Address display.
- o Address input data elements.
- o UK address input.
- o UK address finder.
- o Non UK address input.
- **NHS Number Input and Display:**
- o Input of a fully specified NHS Number.
- o Display of a fully specified NHS Number.
- **Patient Name Input and Display:**
- o Defining the valid values for Patient Name Input and Display
- o Labelling of information, including:
    - Definition of the elements of a Patient Name.
    - Definition of the values for each element.
    - How items of information are to be labelled; this will cover the label text, positioning and any elements of styling required to differentiate labels visually from data.
- o Control layout and structure, in order to achieve:
    - Optimal visibility of the values.

    - Easy recognition of the values in the context of the wider clinical application.
    - Easy recognition of data type requested for input.
    - Reduction of invalid entries.
  - Size of input fields, in order to:
    - Avoid wasting screen space.
    - Ensure optimal display of entire data items.
- **Sex and Current Gender Input and Display:**
  - Defining the valid values for current Sex and Current Gender.
  - Labelling of information, including:
    - Definition of the terms Current Sex and Current Gender.
    - Definition of the values.
    - How items of information are to be labelled; this will cover the label text, positioning and any elements of styling required to differentiate labels visually from data.
  - Control layout and structure, in order to achieve:
    - Optimal visibility of the values.
    - Easy recognition of the values in the context of the wider clinical application.
    - Easy recognition of the data type requested for input.
    - Reduction of invalid entries.
- **Telephone Number Input and Display:**
  - Defining the valid values for Telephone display and input.
  - Control layout and structure, in order to achieve:
    - Optimal visibility of the values.
    - Easy recognition of the values in the context of the wider clinical application.
    - Easy recognition of data type requested for input.
    - Reduction of invalid entries.
    - Optimal size of input fields.

The following aspects are **out of scope:**

- **Address Input and Display:**
  - **Validation of an entered address as a real address –** Techniques to determine whether an entered address is an actual address.
  - **Validation that a given address is that of the stated person –** Techniques to validate whether or not a valid address is that of the stated person.

- **Multi-language applications** – Languages that use right-to-left writing, such as Arabic, the Cyrillic alphabet, such as Russian, or ideograms, such as Japanese.
- **Display styles** – Choice of display font size, background and foreground text colour will affect the readability of addresses, as it will with all other displayed text.
- **Reduced-size form factors** – Handheld devices, such as personal digital assistants (PDAs) and other such small mobile devices.
- **Data storage and transmission** – The guidance relates only to the display layer of a clinical application, and does not prescribe how addresses should be stored. It is assumed that all applications will be capable of transforming an address stored in an arbitrary format into that prescribed by this guidance, without error.
- **Data history and provenance** – The recording of validity dates is left to the designer of the NHS clinical application.
- **Address types** – Entering multiple addresses, such as for office and home, is left to the designer of the NHS clinical application.
- **Method of providing help text and user messages** – There are many ways of providing the user with assistance, such as tooltips, watermarks, FAQ files, and online help.
- **Address picker** – Third-party postcode-based address finders offer a set of candidate addresses.
- **Form design** – Typically, an address will be entered in a form along with other information such as name and email address; the positioning of these and other fields is left to the form designer.

- **NHS Number Input and Display:**

- **Validation** – Conformance to the required format and checksum calculation is defined in the *UK Government Data Standards Catalogue* **{R5}** and is outside the scope of the guidance, as is validation that a given NHS Number is that of the stated person.
- **Multi-language applications** – Languages that use right-to-left writing, such as Arabic, the Cyrillic alphabet such as Russian, or ideograms such as Japanese.
- **Display styles** – Choice of display font size, background and foreground text colour will affect the readability of NHS Numbers, as with all other displayed text.
- **Other patient identification numbers** – Any locally allocated patient identification number used in a hospital's Patient Administration System (PAS) or other systems.
- **Bar code representation** – The display on a wristband of an NHS Number in the form of a bar code.
- **Reduced-size form factors** – The guidance does not cover reduced-size form factors, such as personal digital assistants (PDAs) and such other small mobile devices.
- **Data storage and transmission** – The guidance relates only to the display layer of a clinical application, and does not prescribe how NHS Numbers should be stored. It is assumed that all applications will be capable of transforming an NHS Number stored in an arbitrary format, into that prescribed by this guidance, without error.

- **Data history and provenance** – The recording of dates for when an NHS Number is valid is left to the designer of the NHS clinical application.
- **Form design** – Typically an NHS Number will be entered in a form in which a user is asked to enter additional information such as name and contact address. The guidance does not address form-level aspects such as the positioning of labels, error messages, or how mandatory fields should be displayed.

- **Patient Name Input and Display:**
  - **Data storage** – The guidance does not prescribe the format for storing data that is input or displayed.
  - **Terms of use** – The guidance does not define when an input field or display should be presented within a system.
  - **Form design** – The guidance does not prescribe the correct layout for a form, the navigation around a form, or how these controls should be labelled.

- **Sex and Current Gender Input and Display:**
  - **Data storage** – The guidance does not prescribe the format for storing data that is input or displayed.
  - **Terms of use** – The guidance does not define when an input field or display should be presented within a system or when Current Gender should be used instead of Sex, and vice versa.
  - **Form design** – The guidance does not prescribe the correct layout for a form or the navigation around a form.

- **Telephone Number Input and Display:**
  - **Data storage** – The guidance does not prescribe the format for storing data that is input or displayed.
  - **Terms of use** – The guidance does not define when an input field or display should be presented within a system.
  - **Form design** – The guidance does not prescribe the correct layout for a form or the navigation around a form.
  - **Fax numbers** – The guidance does not prescribe the format for fax numbers.
  - **Data input control types** – The guidance does not specify how the input controls should be labelled, for example, ‘Home number’ or ‘Preferred contact’. The concept of types and context are not explicitly supported. However, the flexible behaviour of the input control and the exclusion of a control-level label enable the developer to reuse the same input control for multiple situations.

4.3. Date and Time Input and Display

The purpose of the guidance is to make display and input of date and time consistent across all clinical systems within the NHS in England. The objective is to increase patient safety by maximising clinical utility and minimising reading and transcribing errors.

Displaying unambiguous information in a consistent format is a core element in ensuring effective patient care. It is vital that healthcare professionals correctly interpret dates and times relating to patient demographics, clinical episodes and planned treatments, among others.

No existing standards (W3C and ISO) provide an entirely unambiguous date display. ISO stipulates that the day and month elements of a date are pairs of numerical values. This presents a risk of date misinterpretation by confusing month for day and vice-versa. W3C reduces this ambiguity by using letters for the month, but it does not specify a format for long dates to complement its short date format.

These 3 design guides specify common user interface aspects of:

- **Date Display** including: General data display guidance; Single precise date display – short date format and long date format; Display of approximate dates comprising only a year or a combination of month and year.
- **Time Display** including: Single precise time display; Approximate (or fuzzy time); Relative times; Time duration; Display of date and time combinations.
- **Date and Time Input** including: Fully specified dates; Fully specified times; Partially specified dates; Approximate dates and times; Arithmetic shortcuts; Date and time durations; Input via free text fields; Input via input controls; Disambiguation of date entries.

The following aspects are **out of scope** for Date and Time Display and Input:

- **Data storage** - The standard relates only to the human interface layer of a software application, and does not prescribe the format for storing data.
- **Styling** – Choice of display font size, background and foreground text colour will affect the readability of all displayed text. These standards will not address general rules for text display. These will be addressed in other guidelines.
- **Reduced-size form factors** – This standard does not cover reduced-size form factors, such as personal digital assistants (PDA's) and other mobile devices.

5. Safety Goals

The Clinical Safety Case rests on three factors:

1. Identification of safety goals and requirements.
2. Evidence which supports safety goals and requirements have been achieved.
3. The safety argument which explains how the evidence can be reasonably interpreted as indicating “acceptable safety”.

The concept of ‘acceptably safe’ is discussed later in the Safety Argument (Section 7).

5.1. Top Level Safety Goal

The top level safety goal for the 9 CUI design guides covered by this *pre-operational* Clinical Safety Case and Closure Report, is that the design guidance is **acceptably safe** for system developers to begin implementing within clinical systems in the NHS in England. The assumption for this safety goal being that this pre-operational safety case is the starting point for the aforementioned system developers’ *operational* safety cases.

5.2. Safety Requirements and Objectives

The following safety goals apply to **all** CUI design guidance:

- CUI design guidance is developed using an assured, fit for purpose development lifecycle.
- The CUI CAPS design guide development lifecycle includes an integrated, fit for purpose safety assessment and management process.
- All identified hazards have been appropriately managed.

Safety specific requirements and general objectives for **each** design guide have been identified as follows:

Patient Banner:

- Reliable and accurate identification of an individual patient record.
- Matching a patient record with:
  - The correct patient, whether present in person or by phone.
  - Other artefacts associated with the patient, for example, samples, letters, or wristbands.
- Displaying core information according to existing standards and guidance and using a minimum data set available to all NHS clinical applications.

- Promoting consistency across the mix of users, NHS clinical applications and care settings.
- Displaying minimum supplementary information to support patient identification or enhance patient care.
- Minimising opportunities where patient-clinician confidentiality and patient privacy may be compromised.
- Minimising opportunities for human error.

Address and NHS Number Input and Display:

- Display information conforming to convention and existing best practice with which clinicians are already familiar, so as to reduce the training requirements of clinical applications.
- Promoting data quality to reduce the occurrence of errors.
- Balancing the need for consistency and commonality across clinical applications with the need to support Independent Software Vendor (ISV) requirements for flexibility.

Patient Name, Sex and Current Gender and Telephone Number Input and Display:

- Display information according to existing standards.
- Minimise opportunities for human error.
- Display sufficient instructional information to support data quality.
- Promote consistency across the mix of users, NHS clinical applications and care settings.
- Ensure reliable and accurate identification of an individual patient record.
- Minimise opportunities where patient-clinician relationships may be compromised.

Date and Time Input:

- Enable dates and times to be entered in a range of formats and using a variety of input methods, including arithmetic shortcuts.
- Support application scenarios where the user needs to enter approximate dates or times.
- Provide the ability to enter either fully specified or partially specified dates.
- Simplify the entry of times by standardising on the 24-hour clock.
- Reduce the likelihood of errors by providing disambiguation of the input, where appropriate.
- A system will be built using a calendar database that recognises the change between Greenwich Mean Time (GMT) and British Summer Time (BST) where appropriate.

Date Display:

- Eliminate confusion between the month and day values.
- Minimise the space required to display dates on a screen.
- Maintain a reading pattern that is natural to users.
- Eliminate opportunities for misinterpreting the date as representing some other data.
- Promote consistency across NHS applications by defining a set of two permissible date formats.

Time Display:

- Enable all time information to be represented explicitly and completely, eliminating the occurrence of ambiguous times.
- Reduce the possibility of misinterpreting the time as a date or other information display.
- Maximise the readability of the time by the use of clear separators between time elements.
- Support application scenarios where the user needs to enter and view an approximate time or duration.
- Promote consistency across NHS applications by providing a small set of valid formats.

6. Patient Safety Assessment

As part of the development of an evolving Clinical Safety Case for CUI design guidance, a Patient Safety Assessment has been completed for each. Section 4.4 (Clinical Safety Approach) described the *high level general* approach CUI takes for PSA's. The *actual* activities which took place for this set of CUI design guidance is described below:

- Primary research – representative end user one to one interviews at their place of work and online (web based) surveys (which have both been geared to generate potential hazards), eg:
    - Patient Banner: A Web based survey of 65 clinicians covering a range of patient identification issues; A further web based survey of 158 NHS clinicians and administrative staff; 12 one to one interviews with a range of healthcare professionals; A plenary discussion with 10 members of the Clinical Spine Applications Design Steering Group (Southern).
    - Address, Patient Name, NHS Number, Sex and Current Gender and Telephone Number Input and Display: A web based survey of 41 respondents from NHS clinicians and administrative staff, Independent Software Vendors (ISVs), community pharmacists and NHS Connecting for Health.
- Secondary (desk) literature research, eg. desk based research project looking at a range of standards, frameworks, literature, information entry web pages and clinical applications.
- Interviews with appropriate Accredited Clinicians (risk).

- Internal review by the CUI CAPS Specific Audience (SA).
- Patient Safety Risk Assessment (PSA) Workshop(s) with end user representation, Subject Matter Experts (SMEs) and nominated Accredited (risk) Clinician(s) / Safety Engineer(s). The main purpose of the PSA workshops was to support hazard identification and risk assessment (see Section 6.1 – Hazard Identification and Risk Assessment).
- National Integration Centre technical review of the CUI design guides.
- ISB assurance and sign off at both Requirement and Draft Standard stage. Draft Standard submission including an end user testing exercise, using a notional clinical application with the CUI design guidance covered by this document adopted within it. (Documents available by request). The Full Standards for all 9 CUI design guides were subsequently approved by ISB either directly at the September 2009 Board meeting, or following the meeting pending satisfactory resolution of actions.
- Hazard Log for each completed and maintained using an appropriate template. All patient safety issues, regardless of where raised, have been added to the Hazard Log for tracking and mitigation. (See Section 6.2 – Hazard Analysis and Risk Management).

6.1. Hazard Identification and Risk Assessment

For Patient Banner and Demographic Inputs and Displays, the initial hazards / risks were identified in a joint PSA workshop on 4 July 2007 at the Department of Health Informatics Directorate London office involving 8 clinicians, 3 CUI CAPS Team Members, 3 user interface designers, 1 Safety Engineer, and 1 Accredited Clinician.

For Date and Time Input and Display, the initial hazards / risks were identified in a joint PSA workshop on 12 July 2007 involving 5 clinicians, 2 technical SME's, 3 user interface designers / CUI CAPS Team Members, 1 Safety Engineer and 1 Accredited Clinician.

The PSA workshops consisted of the following:

1. Preliminary introductions and initial background information provided by CUI CAPS Project Team.
2. Introductory presentations – This elaborated on the objectives of the workshop, how the Patient Safety Assessment is to be created, controlled and finalised, and a concise history of events leading up to the PSA workshop.
3. Hazard identification – Using the Structured What If Technique (SWIFT) to identify / brainstorm hazards, by walking the participants through clinical scenarios of use and displaying click through presentations (PowerPoint slide deck) to mirror the particular UI function(s) being assessed.
4. Hazard assessment – Grading hazards in terms of risk by detailed discussion around consequences, their severity and likelihood, existing safeguards and potential recommendations for mitigation.

To summarise, the patient safety risk assessment approach that was used was:

1. What could go wrong? (Hazard).
2. Why might the hazard occur? (Cause).
3. What are the most likely consequences? (Severity).
4. How likely is the consequence to occur, given the identified hazard? (Likelihood).
5. What are the existing safeguards in place to prevent, reduce or mitigate the patient safety risk?
6. What are the recommendations for mitigation?
7. Grading of risks, using a Risk Matrix. (See Appendix 1).

6.2. Hazard Analysis and Risk Management

Further to the design phase of the CUI CAPS development lifecycle, an evaluation of user interface design work against the known clinical risks was carried out by the CUI Clinical Safety Advisor (CSA) and the CUI CAPS team. This was in preparation for the submission of the CUI design guides as Draft Operational Information Standards to the Information Standards Board for Health and Social Care (ISB) on 3 September 2008. A further evaluation of the Hazard Logs was carried out by the CSA following National Integration Centre (NIC) technical assurance on 5th November 2008 (where issues raised by the NIC were added to the Hazard Log).

A Change Request was authorised in January 2009 to address all outstanding issues. On the proviso that the Change Request was delivered, ISB approved all the CUI design guides as Draft Standards. (Patient Name Input and Display received Conditional Approval). The Change Request work completed in July 2009, upon which a further evaluation of the Hazard Log was carried out by the CSA in order to prepare and submit this report and the Full Standards submissions to ISB.

It is the aim of the CUI CAPS project to manage all risks identified by one of the following options:

1. Mitigate through design, ie. design out so the risk is eliminated.
2. Transfer the risk.
3. For low grade, tolerable risks where designing out is not a viable option, to offer implementation guidance.

All hazards and issues raised to date, as a result of the above processes, have been recorded in a Hazard Log. Section 8 – “Risk Tables” provides a summary of the key risks raised throughout the process which have been assessed as **in scope for CUI CAPS.** Please note, additional risks were raised throughout end to end development, which were either a) designed out during early stages; b) not assessed to be patient safety relevant; or c) not considered in scope for the CUI CAPS project.

Furthermore, as all issues raised by the various assurance processes (eg. ISB, NIC technical assurance) were recorded in the Hazard Log for appropriate tracking, not every hazard or issue has been taken down the cause / consequence / risk grading route. In many cases, where an issue has been raised, CUI CAPS has responded to it immediately without the need for further assessment of the potential risk.

7. Safety Argument

The purpose of this Clinical Safety Case and Closure Report to outline the 'safety argument', ie. the argument (with supporting evidence) that the CUI design guides covered by this document are **acceptably safe** for system developers to **begin** implementing within clinical systems in the NHS in England.

As CUI design guidance is pre-operational, the Clinical Safety Case is also pre-operational. Clinical system developers **must** ensure implementation of CUI design guidance forms part of the operational Clinical Safety Case for each clinical system / release in its specific context of use.

7.1. "Acceptably Safe" and Tolerance Levels

Providing evidence that residual risk has been mitigated to an acceptable level is the ALARP (As Low As Reasonably Practicable) concept – a principle embedded in UK law since 1949. Determining that a risk has been reduced to ALARP involves an assessment of the risk and the cost, time and effort involved establishing controls to mitigate the risk, and a comparison of the two. In the case of clinical safety risks, where deliverables such as CUI design guidance ha been developed in order address existing patient safety objectives, determining ALARP can also include an assessment of the clinical benefits against the residual clinical risk. However, there must be *gross disproportion* between the risk and benefit in order to claim ALARP.

The CUI CAPS project has set acceptable levels and management of clinical risk as follows:

- Risks graded as "Low" or "Safety Integrity Level" (SIL) 1 – 3 are deemed acceptable for CUI design guidance (see the DHID / NPSA Risk Matrix at page Appendix 1). However, given the pre-operational nature of CUI design guidance, **all** risks identified, regardless of their grading, will have been managed in one or more of the following ways:
    - Mitigation by 'clinical risk avoidance', ie. 'design out' of the CUI design guide, in order that the hazard no longer exists.
    - Transfer the risk. Risks requiring transfer in the CUI CAPS project are usually one of three types:

    - Risks which are out of scope for CUI CAPS, however, are relevant for other projects or programmes within the DHID. In the Hazard Log, these types of risks have a ‘Stage’ and ‘Status’ as ‘Transferred’ and ‘Closed’.
    - Risks which are in scope for CUI CAPS, however, mitigation of the risk must take place by those implementing CUI design guidance, eg. system developers or, in some cases, end users of systems, eg. clinicians. In the Hazard Log, these types of risks have a ‘Stage’ and ‘Status’ of ‘Live Risk’ and ‘Transferred’.
    - Risks which are not in scope for CUI CAPS to mitigate (as not in scope for the particular design guide), however, may be a known or unknown risk with current clinical (for example) practice, which is related to the design guide area. Risks of this type have a ‘Stage’ of ‘Invalid’. ‘Status’ could be ‘Closed’, ‘Open’ or ‘Transferred’ depending on the nature of the specific risk and recommendations made by the CUI CAPS project.
  - For low grade, tolerable (as defined above) risks, where avoidance or transfer of the risk is not an option, the ‘Stage’ will be ‘Live Risk’ or ‘Partially Mitigated’ and the ‘Status’ will be ‘Open’.

‘Acceptable safety’ in terms of the overall safety goal for the CUI CAPS project, is that there are **no identified live and open risks above SIL 3 for all CUI design guidance**.

In order to demonstrate this, it is argued the safety case rests on two criteria:

a. Demonstration of adherence to a fit for purpose clinical safety process:

The clinical safety process undertaken by the CUI CAPS Project is consistent with the Department of Health Informatics Directorate (DHID) Clinical Safety Management System (CSMS) and is outlined both at a high level and specifically for each deliverable earlier within this document. See sections 3, 5 and 6.

Please note, some time has elapsed since original development of these particular CUI design guides and the subsequent evaluation of the PSA’s by the (in post since February 2008) Clinical Safety Advisor in preparation for CSG and ISB submissions. As the CSA was not in post at the time of the PSA’s and original development, the safety documentation has been compiled in best efforts based on information available.

b. Mitigation of live risks to as low as reasonably practicable (ALARP):

All risks are argued to have been acceptably mitigated in line with CUI tolerance levels as set (see Section 7.1 - “Acceptably Safe and Tolerance Levels” and Section 8 - “Risk Tables”). To reiterate, the argument for clinical safety outlined within this document is for **pre-operational** use of CUI design guidance.

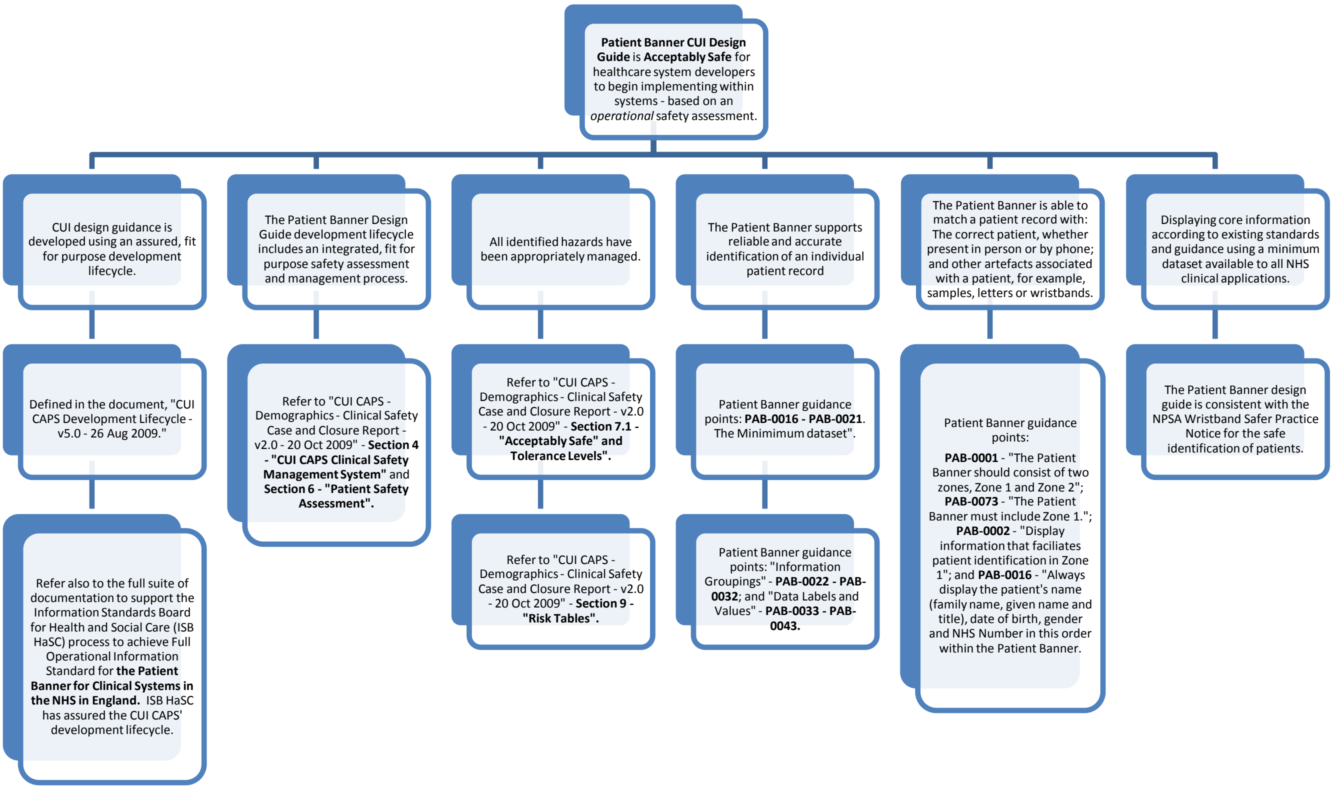
Any healthcare system developer adopting CUI design guidance **must** undertake a further Patient Safety Assessment (PSA) for the implementation (and subsequent deployment) of CUI design guidance within their own healthcare application **based on its specific context of use**. Examples of operational context includes (not exhaustive):

- Setting, eg. primary care, acute secondary care (inpatient and / or outpatient), community, etc.
- End users, eg. doctors, nurses, allied health professionals, administrative and clerical employees, patients, etc.
- Clinical environment, eg. high dependency unit, day surgery, GP consultation, acute psychiatric ward admission, etc.

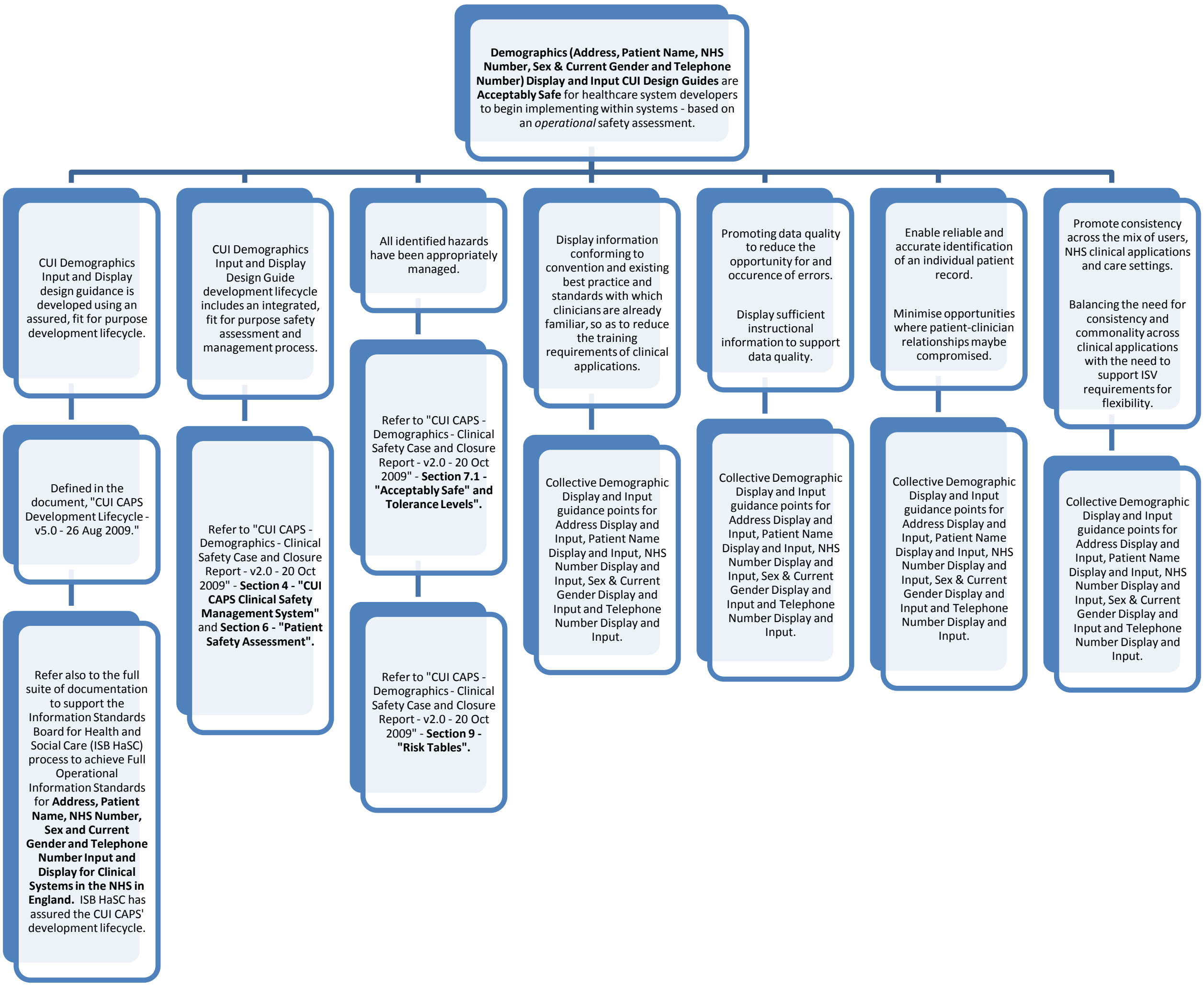
The development lifecycle for CUI design guidance has been adapted in order to ensure that safety risks are proactively and systematically identified and managed throughout the end to end lifecycle. This includes adapting usual project development activities, eg. research, end user consultation, etc. to incorporate patient safety requirements. As such, hazards and risks are often *identified* which fall outside of CUI CAPS' responsibility to *manage.* For example, the pre-operational PSA may highlight a risk which requires mitigation during implementation. Risks of this type will be flagged as having a stage of "Live" or "Partially Mitigated" and a status of "Transferred".

Furthermore, hazards and risks are often identified which fall outside of CUI CAPS' design scope. Such risks can be of relevance to healthcare system suppliers and / or other departments within DHID. Risks of this type will be flagged as having a stage of "Invalid" and status as "Transferred". Therefore, the pre-operational safety documentation referenced and embedded in this document should be the starting point for each *operational* PSA, taking note that there are identified risks where responsibility for mitigation has been 'Transferred' to system developers.

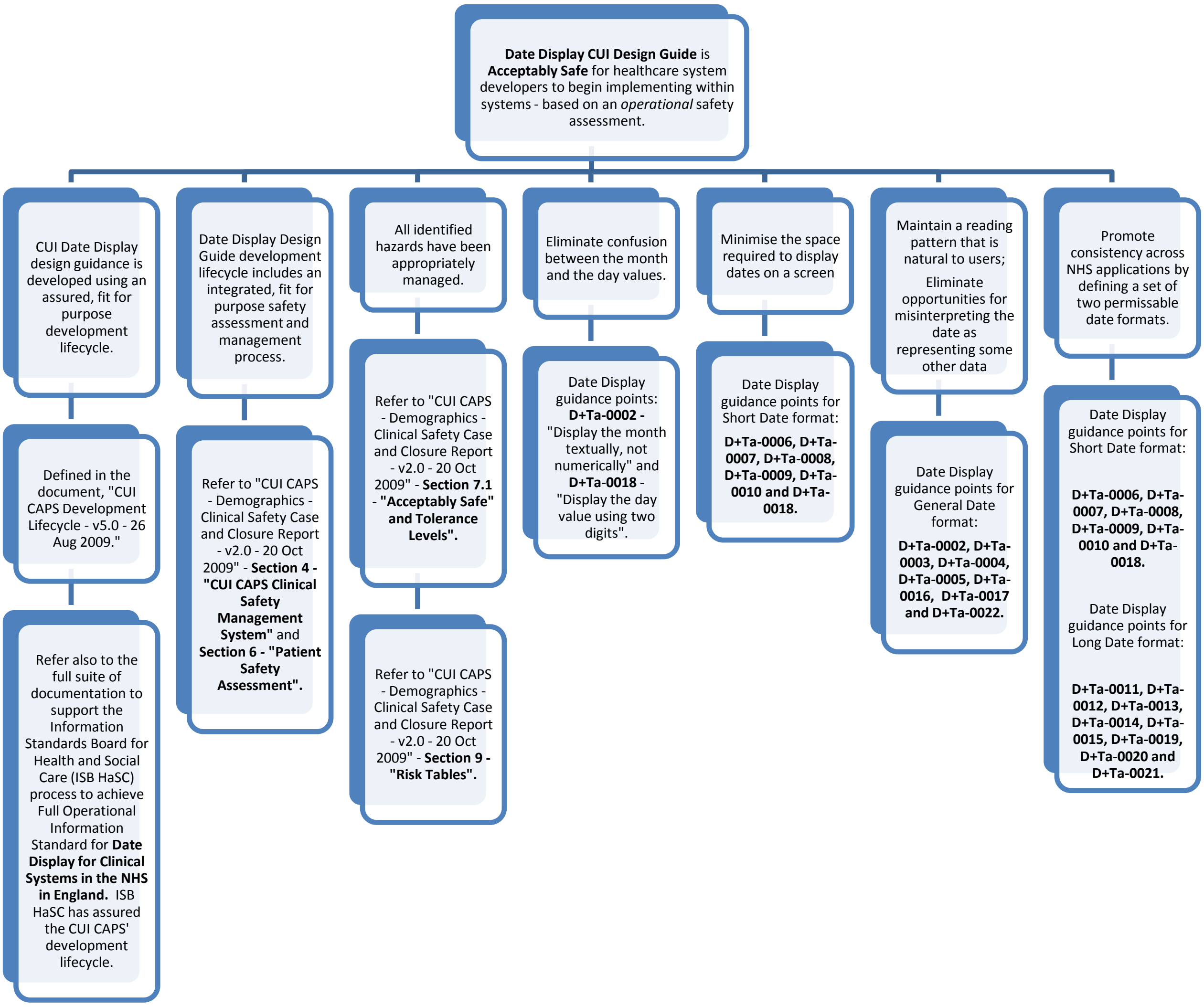
7.2. Patient Banner



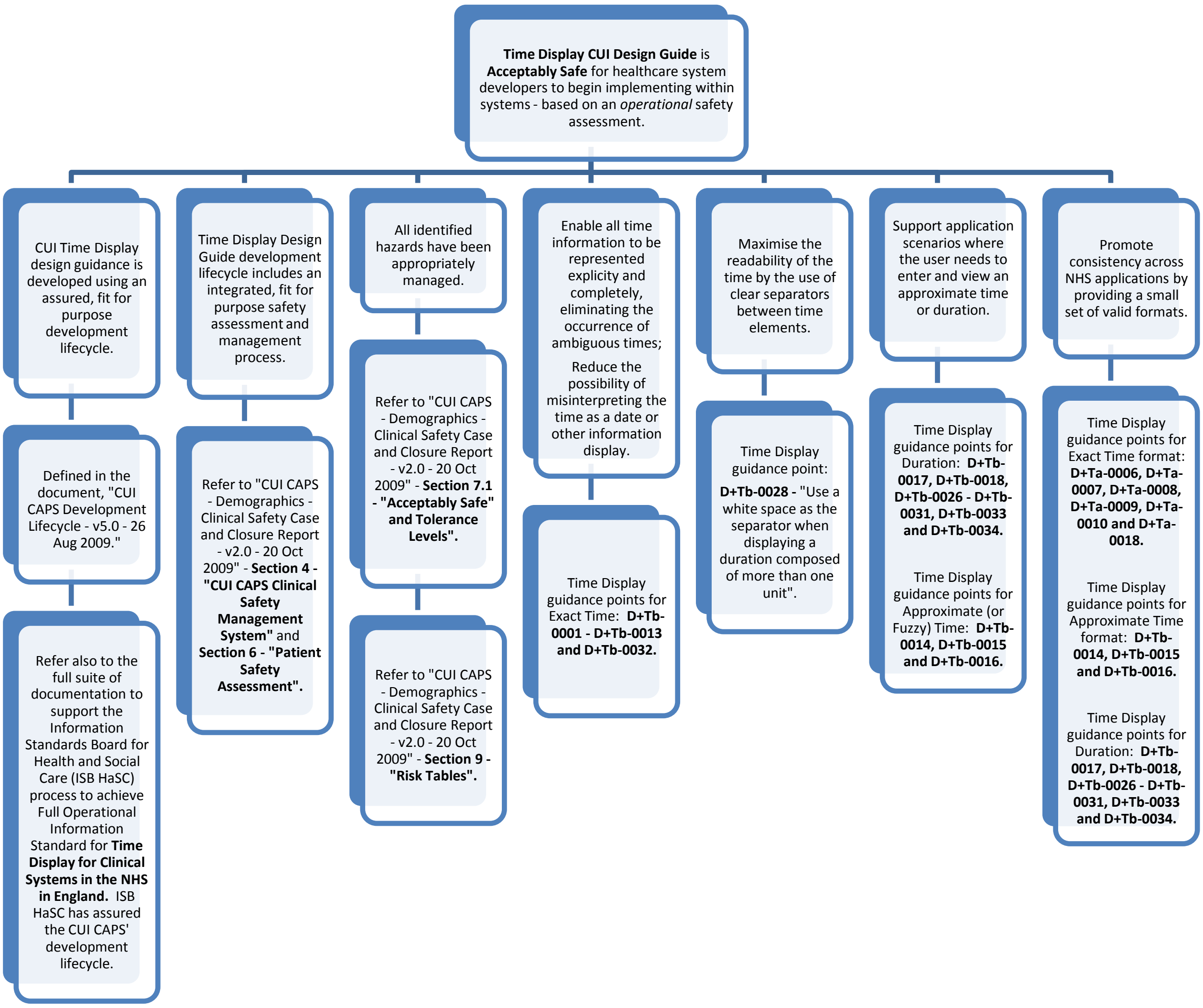
7.3. Demographics Input and Display



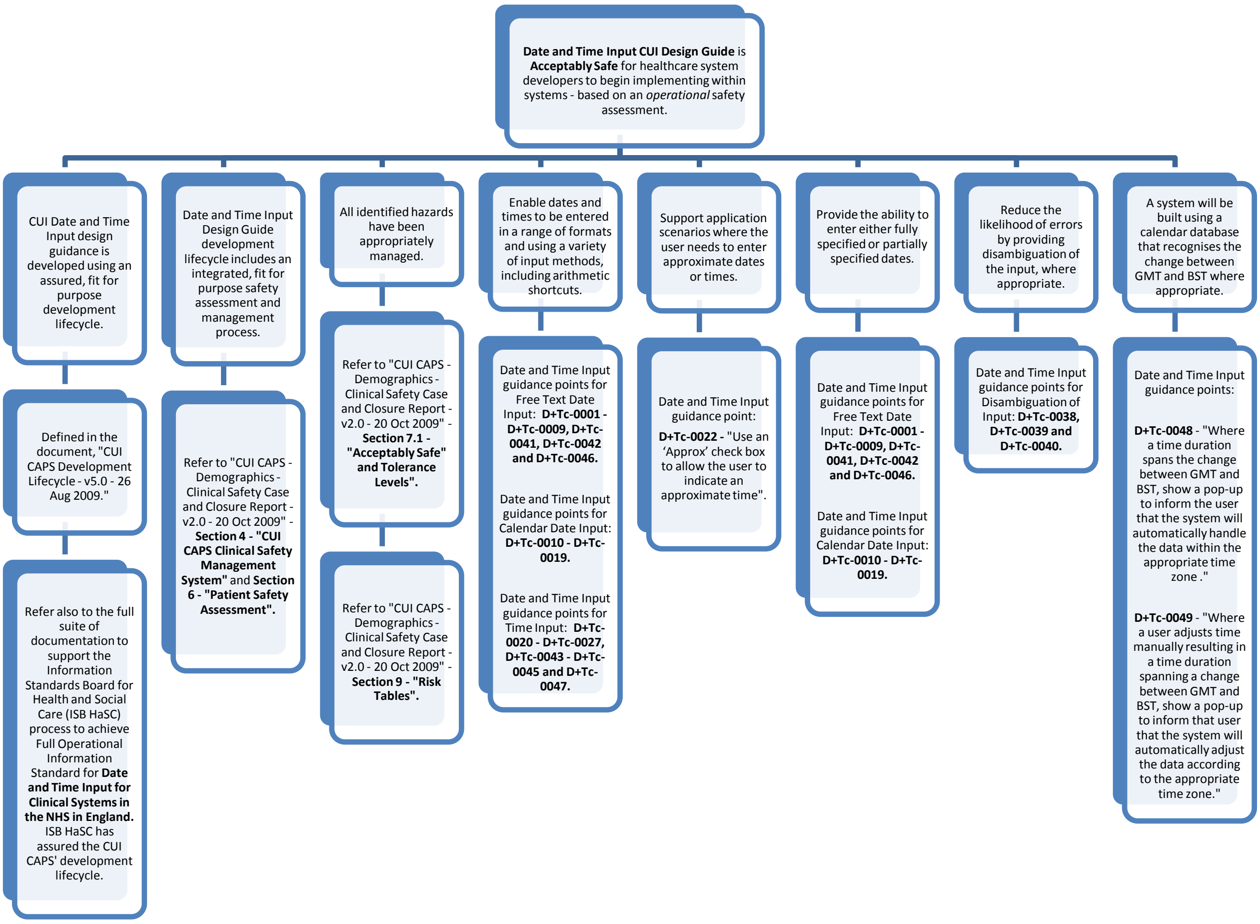
7.4. Date Display



7.5. Time Display



7.6. Date and Time Input



8. Risk Tables

<u>**Note:** Risk Grading is not always completed where hazard has automatically been designed out or the hazard is not strictly a patient safety hazard, but a general issue.</u>

8.1. Patient Banner

No.	Hazard Summary	Cause(s)	Consequence(s)	Risk Grading			Mitigations
				C	L	R	
PAB001	What if clinical judgements are made based on administrative gender and Sex and Current Gender is different?	Sex and Current Gender do not match and gender is shown and not sex.	Inappropriate clinical care for the patient's gender and / or sex.	2	1	2	Mitigated by the CUI Sex and Current Gender design guidance – to be included in the Patient Banner.
PAB002	What if the user confuses given name and family name?	Because there is no clear indication which way round the names appear. (The parts are not labelled).	Care provided to wrong patient.	3	1	3	Mitigated by Mandatory Guidance Point PAB-0053 – "Display the patient's family name in upper case and the patient's given name and title in title case."
PAB004	What if the patient uses a preferred name instead of their given name?	The system does not display their preferred name.	Possible non response from patient as they are used to hearing their given name. Patient frustration.	2	1	2	Mitigated by recommended guidance point PAB-0020 – "Display the Preferred Name where available", and PAB-0037 – "Display the Preferred Name with the label Preferred Name."

PAB006	What if the user misinterprets the age for some other period in time, eg. last time attended?	There is no label for age and there is no visual grouping with Date of Birth.	User confusion.	2	1	2	Mitigated by Mandatory guidance point PAB-0033 – "Precede the data of birth with the label "Born", and PAB-0034 – "When displaying the age of a living patient, place it in parentheses immediately following the date of birth and without a label.
PAB008	What if the user doesn't realise that the patient has multiple addresses / contact details to contact them on?	The system doesn't indicate which address is displayed or that there are additional addresses available (the example only shows one address ("Usual Address" rather than "Postal Address").	Wrong address used. Poor discharge planning. Delay in care. Breach in confidentiality.	3	1	3	Mitigated by PAB-0027 – "Precede the full address with the label "Usual Address", "Temporary Address", or one of the types of temporary addresses, and as defined in PDS. Also recommended in Patient Banner guidance is a link to further contact addresses. Training issue also.

PAB009	What if the user assumes that the address is a contact address and it is not?	The system doesn't indicate which address is displayed and it may be a temporary address.	Wrong address used. Poor discharge planning. Delay in care. Breach in confidentiality.	3	1	3	Mitigated via Mandatory Guidance points: PAB-0025 – "Display as much of the address as possible in a single line, in the title of the first section in Zone 2, displaying an ellipsis to show incomplete display of the address."; PAB-0026 – "Display the full address including the postcode, in the first section of the expanded Zone 2; and PAB-0027 – "Precede the full address with the label "Usual Address", "Temporary Address", or one of the types of temporary address, as appropriate, and as defined in the PDS."
PAB014	What if the user if confused by the ? for Location as it looks like the system does not know where the patient is?	Other zones use numeric's, the display is not consistent and is confusing. Looks like there is a query. Question mark not self explanatory.	User frustration. Delays.	2	1	2	Mitigated via text label.

PAB015	What if the user is confused by the purpose of the Location drop down?	The meaning of Location is ambiguous. No tooltip available.	User frustration. Delays.	2	1	2	Location drop down has been designed out. Mitigated.
PAB021	What if the user doesn't realise that the record is that of a deceased patient and wrongly selects it for the patient presented?	Patient Banner change in colour doesn't appear on different displays.	User frustration.	2	1	2	Mitigated via Mandatory Guidance Points: PAB-0024 – "For a deceased patient, display the date of death and the age at death in Zone 1"; PAB-0038 – "Precede the date of death with the label "Died"; and PAB-0039 – "Precede the age at death with the label "Age at Death".
PAB022	What if the user confuses the date of birth and date of death?	The Date of Death appears as the first field on the screen, where as the user is used to seeing the Date of Birth. Traditionally, DoD is recorded in red. EMIS also shows blue for unregistered patients.	User frustration. Delays.	2	1	2	Mitigated via Mandatory Guidance Points: PAB-0024 – "For a deceased patient, display the date of death and the age at death in Zone 1"; PAB-0038 – "Precede the date of death with the label "Died"; and PAB-0039 – "Precede the age at death with the label "Age at Death".

PAB023	What if, with the large text, the user assumes the date of birth is 14-Jul-1919 (instead of 14-Jul-2019).	Date of birth is truncated.	Inappropriate information about the age of a patient.	2	1	2	Mitigated. Guidance prohibits truncation of information in the Patient Banner.
PAB024	What if the user has the large font switched on and confuses two patients with the same name?	Information is truncated when appearing in large font.	Mis-identification.	3	1	3	Mitigated. Guidance prohibits truncation of information in the Patient Banner.
PAB027	Expansion and collapse of Zone 2 is stated as Mandatory, but not all systems may have this functionality available.	Design.	Implementation consequences. (Not patient safety critical).	1	1	1	The Patient Banner Design Guide has been re-drafted to de-scope mandatory zone to just Zone 1. Zone 2 is now recommended, not mandatory.

8.2. Demographic Display and Input

No.	Hazard Summary	Cause(s)	Consequence(s)	Risk Grading			Mitigations
				C	L	R	
PID003	What if the user cannot see all the address information and selects the wrong patient?	Erroneous address data is hidden OR Only the town is displayed and the user assumes it is the same patient OR The postcode column is at the end of the screen (not associated with the address column) and the user doesn't notice OR There are multiple addresses with a similar start OR The address column is truncated.	Wrong patient selected Delay or incorrect care to patient Incorrect medial history added to wrong patient	2	1	2	Mitigated via ADR-0004: When truncating an address, add an ellipsis to indicate that the address is not displayed in full and, where appropriate, provide a means for the user to access the full address.
PID008	The user can not easily identify the NHS Number	NHS is used as the label, rather than NHS Number OR The number appears as a 10-digit character rather than grouped digits	Delay in care Inefficiency	2	2	2	Mitigated in part by NUM-0002 - NHS Number to be displayed in groups, not a single strand. However, label issue currently not mitigated - guidance does not mandate a label for display of NHS Number. Low risk identified, however, recommendations to be made for training.

PID009	The user can not easily read the NHS Number.	The number appears as a 10-digit character rather than grouped digits.	The number could be transcribed incorrectly Potential patient misidentification Delay in care Lab discarding samples.	2	1	2	Mitigated by NUM-0002 - NHS Number to be displayed in groups, not a single strand.
PID010	The user can not easily read the NHS Number.	Spaces are used in the NHS number and a non-proportional font is used.	The number could be transcribed incorrectly Potential patient misidentification Delay in care Lab discarding samples.	2	1	2	Fonts not in scope for CUI. Also mitigated through NUM-0002: Display the NHS Number as three groups, with a single space included as a separator between groups, as follows: The first group must consist of the first, second and third digits in order; The second group must consist of the fourth, fifth and sixth digits in order; and The third group must consist of the seventh, eighth, ninth and tenth digits in order.
PID012	The user mistakes the NHS Number for the phone number	Brackets around the first three digits of the NHS Number	Delay in contacting the patient/relatives Contacting wrong person Wrong record found (based on phone number).	2	1	2	Mitigated via NHS Number Input Design Guidance as this control now removes brackets (and punctuation).

PID013	The user does not understand the difference between Sex and Gender and changes the patients record	The definition in the rollover is ambiguous OR The user does not know that there is a rollover capability on the labels OR When a roll-over note is presented could the user misattribute the definition to an adjacent field, if the screen (fields) is congested and similar data items are being presented (which is likely)?	User could potentially change the patients information Patient may receive incorrect care or miss out on care (e.g. Screening - if based on Gender value)	2	1	2	Sex and Current Gender CUI design guidance is consistent with business rules for display of sex and gender including definitions.
PID014	The user misinterprets the patient's sex / gender	Numeric value used Abbreviations are unclear	Inappropriate or delayed care	2	1	2	Sex and Current Gender CUI design guidance is consistent with business rules for display of sex and gender including definitions.
PID015	The user is not aware that part of the telephone area code is missing	The 0 is missing as the country code is displayed at the start of the display e.g. +44 (1234) 567890.	Delay in care	2	1	2	New input control only accepts country code for international numbers. Mitigated by design.
PID016	The user dials the wrong number when trying to call the patients mobile	The user misinterprets the brackets as optional digits and excludes them.	Delay in care	2	1	2	Mobile telephone numbers are out of scope for design guidance AND brackets have been removed. Mitigated by design.

PID017	The user mis-dials the patients number	They read the brackets as numbers	Small delay in care, frustration.	2	1	2	Mitigated by design - brackets now removed from phone number in Patient Banner and Telephone Number Display and Input guidance.
PID020	Suffix is defined for input but definition for displaying suffix is unclear - "In most cases, the suffix is not needed". "Some suffixes however are important."						Guidance updated with additional wording to the paragraph to make definition clearer. Issue closed.
PID021	The name input dialogues state that names WILL be recorded in the case the user enters them, but be displayed as per the display rules. What does 'recorded' mean? These are not mandatory requirements stated in the guidance sections.						References to 'recorded' removed. Data storage is also Out of Scope. Issue closed.
PID024	Why are brackets used for telephone numbers when this format is not generally used?						Brackets have now been removed from guidance. Mitigated.
PID026	Some display rules list specific area codes in spacing requirements - telephone numbers are subject to change by regulators and this does not appear to have been given consideration in the document.						Text added to guidance document in recognition that codes will change. Mitigated.

PID027	There are no references to the definition of address in use in other NHS, CFH or SPINE systems. The Patient Name Input and Display document references these in detail.				The only definitions to elements of the address input are contained at 2.2.5.4 to 2.2.5.6. The best way to respond to this point is to collate references into an intro section which mirrors the presentation in the Patient Name guidance. Guidance document updated as such. Issue closed.
PID029	Document provides no guidance, just states that precise lengths can only be given for postcode and county. Why? NHS Data Dictionary lengths for unstructured address is 175 characters and for structured address is 35 chars per line.				Guidance has been updated to refer out to the NHS Data Dictionary, as per advise from the NHS CFH PDS team. Issue closed.
PID031	Out of scope states that the following are out of scope: multi language applications; display styles; other patient identification numbers; bar code representation; reduced size form factors. These items should all be in scope in a guidance document.				Guidance document updated. These items are still out of scope, however, a qualifying reference has been included in this and all other design guides. Issue closed.

PID034	There are observations made for the use of screen readers but no guidance is given.				As per bottom paragraph 2.1.4.2 screen reader audibility patterns vary - therefore any guidance can only relate to the number groupings which as per para 2 prevent large numbers being read out? Wording slightly altered in guidance listed at 2.1.2 to include this number grouping reference. Mitigated.
PID035	The input fields are recommended to contain examples. How will this work when editing existing entries when not all fields have previously been completed? Example data may be confused with actual values.				NID018 suggests that a prompt will be contained within a default field. But there were no guidance points relating to the strength of the example text. Two guidance points have now been added at section 2.1.1 to mitigate this. This has also been included within guidance points in each section. Mitigated.

PID036	The presentation of the 'other' option should be in line and consolidated with existing use of this option.				See section 2.2.1 in guidance. This relates to the parentheses around (Other). This has now been removed. Guidance point NID003 states that the title should have parentheses around to separate. Use of parentheses around just one option within title is confusing. Mitigated.
PID037	Input box examples could extend to direction of the use of UPPER CASE where field input sections are more self explanatory.				See section 2.2.2 which now deals with this adequately. At most perhaps one more usage example re input eg Winstanley or refinement to the wording below figure 11 "When the User moves to the next input cell the Control Displays Family Name text in Uppercase". Mitigated.
PID038 a	Examples of correct usage for extended character lengths should be more live like to avoid misinterpretation.				Agreed - the examples given should be actual examples. Guidance updated. Issue closed.
PID039	Use of desired order of radio buttons is shown but there is no reference to ensure that the tab order of the buttons is adhered to if a keyboard is being used instead of a mouse.				Mitigated via existing Guidance Point CGS-0022 - which is related to 'tabbing'.

PID038	Example use of telephone types does not use brackets around area code.				Addressed with consistency check. Mitigated.
PID039	The recommended display rules use spaces as separator values for area codes, not brackets as stated as mandatory in other sections of the document.				Addressed with consistency check. Mitigated.
PID040	Appendix A lists display rules that would immediately fail if used with PDS data retrieved through TMS messaging.				Appendix A has been reworded to reflect storage as being out of scope.
PID041	MIM telephone format includes telephone type, ie. tel or fax. This would fail provided validation steps. See example below from MIM: <!-- To indicate a home telephone number -> <telecomuse="HP"value="tel:01392251289"/>				Guidance updated to identify fax as out of scope.
PID042	MIM address lengths are defined as NHS Data Dictionary lengths for unstructured address at 175 characters and for structured address at 35 chars per line.				Guidance has been updated to refer out to the NHS Data Dictionary, as per advise from the NHS CFH PDS team. Mitigated.

PID043	What if the Patient Name Input and Display CUI Design Guide uses the labels terms "Given Name" and "Family Name", whereas the NPSA uses the terms "First Name" and "Last Name" in their Safer Patient Identifiers SPN? Caused by the different purpose, scope and end user for the two standards (CUI and NPSA), different research results. Some potential for confusion around the meaning of the terms - does Family Name equate to First or Last Name, for example? Could cause healthcare professional confusion when accessing records of those patients with non conventional names. There is current no standard in the NHS for the terming of name labels, either in systems, data standards, forms, etc. Despite the absence of standard, there are no known patient safety incidents due to the different terms (first name, given name, forename and last name, family name, surname).	3	1	3	Ensure the Patient Name Input and Display CUI Design Guide includes a review of existing data, system and practice standards (including the NPSA SPN) and the name labels terms used for each. However, outline that for the purposes of the design guide, the terms Given Name and Family Name will be used throughout and they are directly synonymous with First Name and Last Name or Forename and Surname. CUI Design Guide end users are healthcare system developers who will need to map data standards to display standards (both of which use Given Name and Family Name) and therefore CUI believe this is the safer and more appropriate approach. The terms themselves however will not be made mandatory in the CUI Design Guide. Tooltips which offer synonyms are also recommended. Risk transferred to healthcare IT developers.

8.3. Date and Time Display and Input

No.	Hazard Summary	Cause(s)	Consequence(s)	Risk Grading			Recommended Mitigations
				C	L	R	
DT003	User enters inappropriate format entry	User misunderstands the meaning of the input mask.	Incorrect data entry	2	1	2	Control does validation that checks for inappropriate entry. Mitigated.
DT004	User doesn't understand "arrow keys" in the pop-up	Not clear what arrow keys mean - could be keyboard arrow keys or the user may be expecting to see spinner arrows. Or should the use they up & down or left & right keyboard arrow keys.	Frustration to user Inefficiency	2	1	2	Instructional text modified. Mitigated.
DT005	User doesn't see that the time is also required.	Date pop-up covers Time input box.	Time is not entered or default time is entered.	2	1	2	Instructional text reduced in size. Mitigated.
DT006	The user thinks that the pop-up is related to a different field	There is no indication to show that the pop-up is related to a field, except that the field has focus.	User doesn't understand what the pop-up relates to.	2	1	2	Instructional text reduced in size. Mitigated.
DT007	The user doesn't understand what is meant by a "spinner" in the time pop-up.	Term is ambiguous / User is unfamiliar with IT terminology.	User doesn't understand the term	2	1	2	Instructional text modified. Mitigated.
DT008	User assumes spinner behaviour follows drop-down behaviour. Drop-down increases with down, where as spinner requires the up arrow.	Standard Windows component behaviour.	User frustration Inefficiency.	1	2	2	This is a risk mitigated by training. This is standard Windows behaviour.

DT009	User assumes that the default time is "now" and enters -10min, but default may be 00:00, which can result in a date change.	Default time is not displayed in the time in the time field - default is not "now" and the arithmetic works from default time.	User frustration - if aware Wrong date entered (if user not checking).	2	2	4	This is a Live Risk (Date and Time controls are separate), however, control and, therefore, mitigation including training lies with the implementing Suppliers and ISVs. Risk to be transferred.
DT010	User assumes date has changed when they enter -20min (minus 20min) at 00:10	System does not automatically recalculate the date.	Wrong date entered.	2	2	4	This is a Live Risk (Date and Time controls are separate), however, control and, therefore, mitigation including training lies with the implementing Suppliers and ISVs. Risk to be transferred.
DT011	User assumes date has not changed when they enter -20min (minus 20min) at 00:10.	System automatically recalculates the date.	Wrong date entered.	2	2	4	This is a Live Risk (Date and Time controls are separate), however, control and, therefore, mitigation including training lies with the implementing Suppliers and ISVs. Risk to be transferred.
DT013	User doesn't notice that control reverts to original content when they enter invalid characters.	User enters invalid characters into the control.	Wrong data entered.	2	2	4	The control does not exhibit this behaviour, it displays an error message and sidesteps this. Mitigated.

DT014	Time control does not behave as user expects	User enters more than 24h in the time control or a period that rolls over to the next/previous day OR Asymmetric linking of boxes - Date should not affect Time, but Time should affect Date.	{Behaviour}.	2	2	4	This is a Live Risk (Date and Time controls are separate), however, control and, therefore, mitigation including training lies with the implementing Suppliers and ISVs. Risk to be transferred.
DT015	User enters the wrong unit into the time field.	User enters +2d in the time control rather than the date control.	{Behaviour}	2	2	4	Mitigated: 'd' invalid within time control due to confusion with date control. New GP added D&T Input at 2.3 + corresponding new GP at 2.1.
DT016	Date and time input boxes are not linked and user assumes that they are	User enters an arithmetic value in the time control and expects the date control to be updated if the time falls onto the next or previous day The date and time control are independent and user expects date to change when they increment time using the spin controls. Visa versa.	Wrong data entered.	2	2	4	This is a Live Risk (Date and Time controls are separate), however, control and, therefore, mitigation including training lies with the implementing Suppliers and ISVs. Risk to be transferred.

DT017	Date and time input boxes are linked and user assumes that they are not	User enters an arithmetic value in the time control and does not expect the date control to be updated if the time falls onto the next or previous day.	Wrong data entered.	2	2	4	This is a Live Risk (Date and Time controls are separate), however, control and, therefore, mitigation including training lies with the implementing Suppliers and ISVs. Risk to be transferred.
DT020	User thinks that the system has added/removed an hour in error and manually changes as they are unaware of daylight savings time (to an incorrect time).	System allows for daylight saving.	Duration may be calculated in error due to daylight savings time – worst case scenario.	2	1	2	Mitigated. See Date & Time Input - section 1.3 and D+TC-0048 and 0049. Two guidance points have been added - one to handle manual time alteration, the second to handle time duration input spanning GMT and BST time change.
DT021	User compensates for daylight savings prior to the system compensation (resulting in compensation twice), i.e. They type in +4h (instead of +3h - allowing for addition of an hour) at 23:00 . And converse for BST to GMT.	System allows for daylight saving.	Duration may be calculated in error due to daylight savings time May result in incorrect medication.	2	1	1	Mitigated. See Date & Time Input - section 1.3 and D+TC-0048 and 0049. Two guidance points have been added - one to handle manual time alteration, the second to handle time duration input spanning GMT and BST time change.

DT022	User does not know whether the time is BST or GMT during that period	2 x 2 o'clock with going back an hour from BST to GMT. (1-2am is repeated for that particular day).	May result in incorrect medication	2	1	1	Guidance document changed to address this hazard. See Date & Time Input - section 1.3 and D+TC-0048 and 0049. Mitigated.
DT023	User enters "non-existent" time, e.g. 2:30 on the day that GMT becomes BST	User changes time to be 02:30 and then changes the date to the day that daylight savings occurs. (Moving from GMT to BST).	May result in incorrect medication	3	1	3	Guidance document changed to address this hazard. See Date & Time Input - section 1.3 and D+TC-0048 and 0049. Mitigated.
DT024	The time periods mean different things to different people and different systems.	User does not know the time periods of Morning, Afternoon, Evening and Night.	Confusion over appointment organisation	2	1	2	Time Periods removed. Mitigated.
DT025	The user enters a time period when an exact time is required	User does not know that they can enter a numeric time once they are on the morning/afternoon/etc cycle.	Inefficiency	2	1	2	Time Periods removed. Mitigated.
DT026	User enters an exact time when a time period is required (for multiple patients).	The ability to enter time periods is hidden	Inefficiency Patient frustration	2	1	2	Time Periods removed. Mitigated.
DT027	The user types "e" for the start of "eight" and auto completes to evening	User does not know that they can enter a numeric time once they are on the morning/afternoon/etc cycle.	Inefficiency	2	1	2	Time Periods removed. Mitigated.

DT028	Time periods are inconsistently interpreted	Time periods are not defined on time entry box (pop-up box needed informing the user).	Wrong time period used, confusion	2	1	2	Time Periods removed. Mitigated.
DT028	The user does not realise the significance of a date (For example, discharge date would be unlikely to occur on a weekend - if it has, it may be due to a self-discharge or the day may affect when the follow up appointment is scheduled).	The user can not see the day of the week.	Potential, delay in follow up Delay in care Inefficiency.	2	1	2	Mitigated - Values for display of days of week now standardised.
DT029	Difference between 4w and 1m.	Does 1m = 4 weeks or calendar month. Outlook does not allow arithmetic in months.		2	1	2	Behaviour is "+*n*m" simply jumps to today's date in appropriate month, i.e. **+1m** from today would land on 19/3/2009. *(There is an exception, ie, what happens when the date is Jan 30th and you +1m)*. Week arithmetic behaves as you'd expect, i.e. it jumps whole weeks. Mitigation is familiarity and training. ISV could also add supporting text/tooltips etc on screen if necessary. Risk transferred.

DT029	The user misinterprets the date.	Bold of 9 looks like an 8 OR They misread the leading 0 / find it difficult to read.	Numbers are transposed Missed appointment Delay in care.	2	1	2	Recognised, tolerable risk. Readability should conform to accessibility level.
DT030	The patient is confused about the date of their appointment on the letter.	The day of the week is not included on patient correspondence.	Patient may not turn up Delay in care	2	1	2	Mitigated via D+Ta-0016: When displaying the day of the week, use one of the following abbreviations: Mon, Tue, Wed, Thu, Fri, Sat and Sun.
DT032	The user uses the scroll wheel to scroll the page but actually scrolls the time using the spinner controls, entering an incorrect time.	The control allows operation through the scroll wheel AND It isn't clear that the focus is on the time control.	Incorrect time/date input Incorrect patient record Inappropriate care.	2	1	2	To be mitigated via training. However, this is a live risk.
DT034	The user does not understand and misinterprets what is written in the Duration box.	Coding 1d 3hrs 4min is not intuitive to the user.	Inappropriate care.	2	1	2	Duration units standardised and rationale clarified. Mitigated.
DT035	The user misreads the code in the Duration box (glances and doesn't read the units)	Scaling is inconsistent on the screen, time includes seconds OR They do not read the units and expect to see hrs, min and seconds (May be better to supplement display with a graphical representation?).	Inappropriate care.	2	1	2	Duration units standardised and rationale clarified. Mitigated.

DT036	The user can not see the full duration by glancing at the box and misinterprets it.	The duration is so long that it exceeds the text box, for example, 5y 6m 10d 12hrs 36min 10sec.	Inappropriate care.	2	1	2	Duration units now use one character fewer. Mitigated.
DT037	The user wants to enter a short duration of time.	The time duration is under a minute.		1	1	1	Guidance states you can enter seconds as single digits. Mitigated.
DT038	The user enters or edits the duration, which does not match the start and finish dates and times, and expects the start time to be updated, (or the finish date).	The system does not indicate whether the changing the duration will update the end time or the start time and changes the wrong one.	Inappropriate care.	2	1	2	The guidance covers this event as this is dependant on context of implementation. Mitigated.
DT040	The user enters 26h in the duration box expecting the start date to update (and it doesn't).	The system does not update the date OR The system does not indicate whether the duration box will update the end date or the start date.	Inappropriate care.	2	1	2	Guidance does not prohibit this type of implementation as ISV discretion. They are separate controls and the ISV can manage their interaction. Risk transferred to ISVs.

DT041	The user enters a duration that knocks the time over midnight and expects the date to be updated (and visa versa).	The system does / does not update the date AND does not clearly indicate this.	Inappropriate care. Inaccurate record.	2	2	4	Live risk – to transfer to ISV's. Date and Time controls must be linked where appropriate.
DT042	The user enters an approximate duration which they intended to be approximate, which is recorded with undue accuracy	The user enters a duration with one of the approximate boxes on the time control checked OR The user approximates one time and not the other and calculates the duration OR The user enters 1-2d in the duration box (see below - fractions need investigation) AND The system does not indicate that the duration is calculated/ entered as an approximate period.	Inaccurate record Inappropriate diagnosis (i.e. Pregnancy).	2	1	2	Time display guidance has an add in that approx can be applied to duration if start date or end date is approx. Mitigated.
DT043	Fractions and clinician shorthand	1/2 could be half, 1 to 2, 1 or 2, 1/7 could be 1 day 1/52 could be 1 week 1/12 = 1 month 1/24 = 1 hour 24/7 = 24hrs 7 days a week 8/40 (pregnancy).					Out of scope for the work we undertook. ISV's can implement this behaviour if required in an implementation. The control does not interpret this kind of shorthand in the way that is indicated in column L. It was out of scope for the work. It's all valid stuff but, at the moment,1/52 means January 1952 etc. Issue transferred.

DT044	Decimal points and punctuation.	1.5 days (one and a half days) etc Colons.					Out of scope for the work we undertook based on requirements. ISV's can implement this behaviour if required in an implementation. Risk transferred.
DT045	The user selects the default option when the disambiguation pop-up is displayed (either with mouse or keyboard) rather than the correct date.	Pop-up fatigue (PuF)	Entry of wrong date Incorrect patient record Inappropriate care.	2	1	2	Disambiguation dialogue now only appears for ambiguous dates. Mitigated.
DT048	Disambiguation of two-digit years?		A user might enter a DOB using 2 digit year format it is considered unlikely that a date range would be entered in 2 digit format.	2	1	2	This is a known risk, but tolerable. Likelihood of a clinician using a 2 digit format as per the example is unlikely. Although accepted a user might enter a dob using 2 digit year format it is considered unlikely that a date range would be entered in 2 digit format.
DT049	"Enter 24 hour clock format" label next to the time box.						Mitigated by the presence of a 24 hr clock tooltip (see D+Tc001) and an additional guidance point regarding indication of 24 hour clock use (D+Tc001.).

DT050	The user enters the time using the 12hr clock format, into what they think is a 12hr clock (types 5:00 meaning 5pm) but it is actually 24hr clock. Meaning their time is mis-recorded.	The system displays 05:00 instead of 17:00 as it is a 24hr clock.	Entry of wrong time Incorrect patient record Inappropriate care.	2	1	2	Control now uses 24hour clock only. Mitigated.
DT051	The user misreads or miscalculates a time displayed in 24hr format (i.e. Reads 15:00 hours and thinks its 5pm, or sees 16:00 and deducts 12hrs and gets 6am instead of 4am).	The system only displays 24hr clock - which can be misinterpreted (Would like to see 24hr followed by 12hr clock).	Inappropriate care	2	1	2	Control now uses 24hour clock only. Mitigated.
DT052	The user inappropriately avoids checking Approx.	It is quicker for the user not to tick it OR They want to avoid the "stigma" of admitting to approx/late times (e.g. administration of drugs) OR The system doesn't make the definition of Approx clear OR The user does not understand the term OR The user does not know how and when to use it.	Inaccurate patient record (times look to be exact) Inappropriate drug administration				We provide the ability to enter approx. The use of this is down to the culture of local work practice and governance. This is stated in guidance. Mitigated.

DT053	The user inappropriately checks Approx.	The system doesn't make the definition of Approx clear OR The user does not understand the term OR The user does not know how and when to use it OR The user generalises on time. They use the Approx box all the time, so enter a very general time (i.e. 3:30 and then ticking approx).	Inaccurate patient record Inappropriate drug administration.	3	1	3	This will need to be dealt with by implementing ISVs and training. Transferred risk.
DT054	What if the user misinterprets the month July for June (and vice versa) and January for June (and vice versa)?	Because Date Display Design Guidance mandates months are displayed Jan, Jun and Jul, ie. the first three letters of the month - and could be misinterpreted as they are similar looking to each other.	Misinterpretation of dates - could lead to mis-management, missed appointments, small potential for mistreatment.	3	1	3	Known, tolerable live risk.
DT055	The guidance should also reference any areas where the use of default dates may be input or assumed within systems as being out of scope.						Included in out of scope. Also mitigated via section - 1.2.2. (Date Display) - " Default dates - This guidance only relates to the input of data and not to any default dates assumed by developers."

DT056	The Design Guide entry - Time Display - document does not define its scope.						Mitigated - document updated and restructured to make scope clear. (Time Display - section 1.2.1).
DT057	There is no guidance for the display of milliseconds for example from an audit log - databases and messaging systems often hold time values down to millisecond level.	Display of milliseconds has not been dealt with as there have been no user scenarios identified where the clinician user required time to this level of granularity.					Out of scope. The inclusion of additional guidance around milliseconds will require additional research into audit presentation. Would consider as part of future work (via maintenance and update).
DT058	The guidance of time duration for days, weeks, months and years seems incomplete and there are no examples showing definition of the year value.						Mitigated via changes made to guidance as follows: Time Display section 2.3.2 and Date and Time Input section 2.5.2.
DT060	Recommendations for the use of whole numbers should be 60 minutes not 90, as the document shows 90 minutes being represented as 1 hour and 30 minutes.						Change made to D+TB-0018. Mitigated.

DT061	The display requirements of the days of the week are not specified in the guidance sections of the document or in examples for long date format and whether the full name of the day could be used in the long date format.						Currently out of scope. To be addressed as part of maintenance and update of the design guide once approved as a standard.

APPENDIX 1 – DEPARTMENT OF HEALTH INFORMATICS DIRECTORATE RISK MATRIX

At the time of conducting the PSA’s, the following Risk Matrix was used, as per the CSMS at that time. Therefore, risks initially graded comply with this scoring system.

Likelihood (L)							
Greater than 1 error per day per HP	7	Medium SIL2	High SIL3	High SIL4	High SIL4	High ‘u’	High ‘u’
An error between once a week or once a month per HP	6	Medium SIL1	Medium SIL2	High SIL3	High SIL4	High SIL4	High ‘u’
An error between once a month or once a year per HP	5	Low ‘a’	Medium SIL1	Medium SIL2	High SIL3	High SIL4	High SIL4
An error once a year to once in 10 years per HP	4	Low ‘a’	Low ‘a’	Medium SIL1	Medium SIL2	High SIL3	High SIL4
An error once in 10 to once in 100 years per HP (once in HP lifetime)	3	Low ‘a’	Low ‘a’	Low ‘a’	Medium SIL1	Medium SIL2	High SIL3
An error once in 100 to once in 100 years per HP (once in HP practice lifetime)	2	Low ‘a’	Low ‘a’	Low ‘a’	Low ‘a’	Medium SIL1	Medium SIL2
An error less than once in a 1000 years per HP (once nationally per year across all HPs)	1	Low ‘a’	Low ‘a’	Low ‘a’	Low ‘a’	Low ‘a’	Medium SIL1
		A	B	C	D	E	F
		Negligible / Very Low	**Low**	**Moderate**	**Severe**	**Major / Fatal**	**Catastro-Phic**
Patient Safety Risk Matrix		Minimal extra observation or very minor treatment, and causes minimal harm to a patient.	Extra observation or minor treatment, and causes minimal harm to a patient.	Further treatment, possible surgical intervention, cancelled treatment, or transfer to another area, and which causes short-term harm to a patient.	Permanent or long-term harm to a patient.	A patient fatality	> 3 fatalities. > 10 “Severe” > 100 “Moderate” > 1,000 “Low” > 10,000 “negligible /very minor” harm.

Consequence (C)

Adapted from ‘Risk Matrix Guidance for Patient Safety Risk Assessments’ by the NPSA

However, in the time elapsed since the original PSA's, the following Risk Matrix has been introduced to the CSMS and, as such, re-graded risks in the Hazard Logs comply with this scoring system instead.

LIKELIHOOD	**Certain**	**5**	**5 (Moderate)**	**10 (Significant)**	**15 (High)**	**20 (High)**	**25 (High)**
	Likely	**4**	**4 (Moderate)**	**8 (Significant)**	**12 (Significant)**	**16 (High)**	**20 (High)**
	Moderate	**3**	**3 (Low)**	**6 (Moderate)**	**9 (Significant)**	**12 (Significant)**	**15 (High)**
	Unlikely	**2**	**2 (Low)**	**4 (Moderate)**	**6 (Moderate)**	**8 (Significant)**	**10 (Significant)**
	Rare	**1**	**1 (Low)**	**2 (Low)**	**3 (Low)**	**4 (Moderate)**	**5 (Moderate)**
Department of Health Informatics Directorate			**1**	**2**	**3**	**4**	**5**
			Minor	**Moderate**	**Serious**	**Major**	**Critical**
RISK MATRIX			**CONSEQUENCE**				

Status		**Hazard Rating**	
Open		**Likelihood**	
Closed		Definition	**Score**
Transferred		Rare	1
		Unlikely	2
Hazard Rating		Moderate	3
Consequence		Likely	4
Definition	**Score**	Certain	5
Minor	1	**Rating**	**SIL**
Moderate	2	Low	1 - 3
Serious	3	Moderate	4 - 6
Major	4	Significant	8 - 12
Critical	5	High	15 - 25